

TORINO CASELLE, Italy

WMO: 160590

Lat: 45.201N

Lon: 7.650E

Elev: 989

StdP: 14.18

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.9	25.2	8.7	8.9	34.4	12.4	10.7	33.4	19.5	51.9	14.6	47.2	3.2	300	0.400

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.0	88.1	72.4	86.0	71.1	84.0	69.9	74.8	84.3	73.3	82.7	71.9	80.5	5.2	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.9	122.3	79.1	70.1	114.9	78.1	68.4	108.3	77.3	38.9	84.6	37.5	83.2	36.2	80.5	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.4	11.3	9.5	18.6	92.0	4.1	2.9	15.6	94.1	13.2	95.8	10.9	97.4	7.9	99.6		
			16.7	76.7	3.7	2.1	14.1	78.2	11.9	79.4	9.9	80.6	7.2	82.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.5	37.3	40.3	48.3	54.5	62.3	69.9	73.8	72.7	65.2	56.4	45.9	38.4
	DBStd	13.92	4.98	5.78	6.05	5.75	5.38	5.49	4.34	4.58	5.04	5.61	5.45	5.24
	HDD50	1311	396	274	103	21	0	0	0	0	0	14	142	361
	HDD65	4248	860	692	518	317	116	17	2	3	57	268	572	826
	CDD50	3317	0	2	50	156	382	597	737	704	456	213	20	0
	CDD65	781	0	0	0	2	33	164	274	242	63	3	0	0
	CDH74	5943	0	0	4	25	243	1229	2239	1826	360	17	0	0
	CDH80	1604	0	0	0	2	34	325	659	532	50	2	0	0
Wind	WSAvg	3.9	3.2	3.7	4.4	4.7	4.6	4.5	4.6	4.4	3.9	3.2	3.3	3.0
Precipitation	PrecAvg	35.2	1.5	2.1	3.0	3.5	4.6	4.1	2.5	2.9	3.0	3.6	2.6	1.9
	PrecMax	63.7	6.0	9.6	7.3	10.1	10.9	6.9	9.9	9.2	9.0	11.2	9.0	8.0
	PrecMin	11.0	0.0	0.0	0.0	0.0	0.3	1.5	0.0	0.1	0.1	0.0	0.1	0.0
	PrecStd	11.3	1.5	2.4	2.1	2.5	3.3	1.7	2.2	2.1	2.6	3.2	2.4	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.2	64.5	73.2	77.1	84.1	90.0	91.1	91.8	84.4	75.6	64.5	57.3
		MCWB	46.4	51.2	54.9	60.6	66.3	72.7	73.7	74.4	68.1	61.8	52.7	45.7
	2%	DB	53.3	57.6	67.6	72.8	79.7	86.4	88.0	88.0	80.9	71.6	60.5	52.1
		MCWB	43.6	46.8	52.7	58.1	64.9	71.8	72.4	72.4	66.9	60.7	51.6	44.3
	5%	DB	49.1	53.9	64.0	69.6	76.7	84.0	86.0	85.6	78.5	68.4	57.0	49.7
		MCWB	41.4	44.1	51.4	56.5	63.1	70.3	71.1	71.0	65.2	59.1	50.6	43.6
	10%	DB	46.1	50.4	60.4	66.2	73.3	80.8	84.0	82.7	75.3	65.9	54.1	46.6
		MCWB	40.1	42.5	49.5	54.6	61.4	68.6	70.1	69.2	63.8	57.9	49.5	42.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.4	51.2	56.3	62.2	69.1	76.9	77.1	76.4	71.3	64.6	56.4	49.7
		MCDB	53.7	61.2	67.6	74.0	80.2	85.7	85.1	87.6	78.5	71.2	61.4	52.1
	2%	WB	44.8	47.9	54.1	59.5	66.1	73.8	74.8	74.4	69.2	62.8	54.0	46.6
		MCDB	49.5	55.1	64.2	69.8	77.1	82.7	83.9	84.6	77.2	68.3	57.5	50.0
	5%	WB	42.5	45.6	52.2	57.5	64.3	72.0	73.3	72.8	67.5	61.3	52.5	44.3
		MCDB	46.8	51.1	61.4	67.3	74.3	80.7	82.9	82.1	75.3	66.3	55.5	47.3
	10%	WB	40.5	43.4	50.4	55.7	62.6	70.1	71.9	71.3	65.7	59.7	50.7	42.6
		MCDB	44.3	48.1	57.9	64.2	71.3	78.9	80.8	79.6	73.0	64.3	53.7	45.6
Mean Daily Temperature Range	5% DB	MDBR	15.2	16.6	18.4	17.1	16.7	17.1	18.0	17.7	17.0	15.0	13.9	14.6
		MCDBR	21.5	23.4	25.3	23.0	21.4	19.9	19.9	20.7	20.7	19.2	18.6	19.4
		MCWBR	13.9	14.4	14.1	12.4	10.7	9.5	9.7	9.3	9.2	10.5	12.1	13.0
	5% WB	MCDBR	18.1	20.2	21.7	19.8	18.9	18.0	17.7	18.5	17.3	15.5	13.4	15.0
		MCWBR	12.8	13.3	12.9	11.2	10.2	9.3	9.4	9.4	8.5	9.2	9.5	11.0
Clear-Sky Solar Irradiance	taub	0.309	0.337	0.385	0.436	0.443	0.457	0.442	0.435	0.422	0.397	0.345	0.303	
	taud	2.406	2.330	2.224	2.115	2.164	2.176	2.222	2.249	2.252	2.291	2.391	2.434	
	Ebn at Noon	253	266	266	261	263	259	262	259	251	239	236	244	
	Edh at Noon	23	30	39	47	46	46	43	41	37	31	24	21	
All-Sky Solar Radiation	RadAvg	521	784	1142	1409	1653	1946	2005	1739	1333	849	515	439	
	RadStd	61	99	141	160	141	141	137	105	101	105	78	56	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.95	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+189	+121
(47)	Regional (63 neighbors)	+0.54	N/A	N/A	+1.15	N/A	+0.43	N/A	N/A	+108	+32

Nomenclature: See separate page